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4.2.3 Preparation of Deuterium:

Electrolysis of heavy water:

Normal water contains 1.6 x 10^{-4} percentage of heavy water. The dissociation of protium water (H_2O) is more than heavy water (D_2O). Therefore, when water is electrolysed, hydrogen is liberated much faster than D_2. The electrolysis is continued until the resulting solution becomes enriched in heavy water. Further electrolysis of the heavy water gives deuterium.

$$2\,D_2O \xrightarrow{\text{Electrolysis}} 2\,D_2 + O_2$$

4.2.4 Preparation of Tritium:

As explained earlier the tritium is present only in trace amounts. So it can be artificially prepared by bombarding lithium with slow neutrons in a nuclear fission reactor. The nuclear transmutation reaction for this process is as follows.

$$^{6}_{3}Li + ^{1}_{0}n \rightarrow ^{4}_{2}He + ^{3}_{1}T$$

4.3 Properties of Hydrogen

4.3.1 Physical Properties:

Hydrogen is a colorless, odorless, tasteless, lightest and highly flammable gas. It is a non-polar diatomic molecule. It can be liquefied under low temperature and high pressure. Hydrogen is a good reducing agent. Various physical constants of hydrogen molecule are listed in Table 4.1.

4.3.2 Chemical Properties:

Hydrogen reacts with oxygen to give water. This is an explosive reaction and releases lot of energy. This is used in fuel cells to generate electricity.

$$2\,H_2 + O_2 \rightarrow 2\,H_2O$$

Similarly, hydrogen also reacts with halogens to give corresponding halides. Reaction with fluorine takes place even in dark with explosive violence while with chlorine at room temperature under light. It combines with bromine on heating and reaction with iodine is a photochemical reaction.

$$H_2 + X_2 \rightarrow 2\,HX \quad (X = F, Cl, Br \,\&\, I)$$

In the above reactions the hydrogen has an oxidation state of +1. It also has a tendency to react with reactive metals such as lithium, sodium and calcium to give corresponding hydrides in which the oxidation state of hydrogen is -1.

$$2\,Li + H_2 \rightarrow 2\,LiH$$

$$2\,Na + H_2 \rightarrow 2\,NaH$$

These hydrides are used as reducing agents in synthetic organic chemistry. It is used to prepare other important hydrides such as lithium aluminium hydride and sodium boro hydride.

$$4\,LiH + AlCl_3 \rightarrow Li[AlH_4] + 3\,LiCl$$

$$4\,NaH + B(OCH_3)_3 \rightarrow Na[BH_4] + 3\,CH_3ONa$$

Hydrogen itself acts as a reducing agent. In the presence of finely divided nickel, it adds to the unsaturated organic compounds to form saturated compounds.

$$HC \equiv CH \xrightarrow{Ni/H_2} H_2C = CH_2 \xrightarrow{Ni/H_2} H_3C - CH_3$$